

Home Work 4
II B. Tech. (Common)
Mathematics-II (NMCI102)

Notations:

$M_{m \times n}(\mathbb{F})$: The set of all matrices of order $m \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

$M_n(\mathbb{F})$: The set of all matrices of order $n \times n$ with entries from $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

$P(\mathbb{R})$ or $\mathbb{R}[x]$: The set of all polynomials in one variable with real coefficients.

$P_n(\mathbb{R})$ or $\mathbb{R}_n[x]$: The set of all polynomials of degree at most n in one variable with real coefficients.

1. Find a basis and the dimension of the following vector spaces.

(a) $\mathbb{R}^n$ over $\mathbb{R}$ (b) $\mathbb{C}$ over $\mathbb{R}$ (c) $P(\mathbb{R})$ over $\mathbb{R}$ (d) $M_{m \times n}(\mathbb{R})$ over $\mathbb{R}$

(a) $\mathbb{R}^n$ over $\mathbb{R}$.

We have

$$\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) \mid x_i \in \mathbb{R}\}.$$

Consider the standard unit vectors

$$e_1 = (1, 0, \dots, 0), \quad e_2 = (0, 1, \dots, 0), \quad \dots, \quad e_n = (0, 0, \dots, 1).$$

Every vector $(x_1, \dots, x_n) \in \mathbb{R}^n$ can be written as

$$(x_1, \dots, x_n) = x_1 e_1 + \dots + x_n e_n.$$

Thus $\{e_1, \dots, e_n\}$ spans $\mathbb{R}^n$. If

$$a_1 e_1 + \dots + a_n e_n = 0,$$

then $a_1 = \dots = a_n = 0$, so the set is linearly independent.

$$\boxed{\text{Basis} = \{e_1, \dots, e_n\}}, \quad \boxed{\dim_{\mathbb{R}}(\mathbb{R}^n) = n}.$$

(b) $\mathbb{C}$ over $\mathbb{R}$.

Every complex number can be written as

$$z = a + bi, \quad a, b \in \mathbb{R}.$$

Thus

$$z = a(1) + b(i).$$

Hence $\{1, i\}$ spans $\mathbb{C}$ over $\mathbb{R}$.

If

$$a \cdot 1 + b \cdot i = 0,$$

then $a + bi = 0$, which implies $a = 0$ and $b = 0$. Thus the set is linearly independent.

$$\boxed{\text{Basis} = \{1, i\}}, \quad \boxed{\dim_{\mathbb{R}}(\mathbb{C}) = 2}.$$

(c) $P(\mathbb{R})$ over $\mathbb{R}$. We have

$$P(\mathbb{R}) = \{a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n \mid n \geq 0\}.$$

Every polynomial can be written as

$$a_0(1) + a_1(x) + a_2(x^2) + \dots + a_n(x^n).$$

Thus

$$\{1, x, x^2, x^3, \dots\}$$

spans $P(\mathbb{R})$.

These monomials are linearly independent, since a polynomial is zero if and only if all its coefficients are zero.

$$\boxed{\text{Basis} = \{1, x, x^2, x^3, \dots\}}, \quad \boxed{\dim_{\mathbb{R}}(P(\mathbb{R})) = \infty}.$$

(d) $M_{m \times n}(\mathbb{R})$ over $\mathbb{R}$.

For each pair (i, j) , define E_{ij} to be the $m \times n$ matrix whose (i, j) -entry is 1 and all other entries are 0. Any matrix $A = (a_{ij})$ can be written as

$$A = \sum_{i=1}^m \sum_{j=1}^n a_{ij} E_{ij}.$$

Thus the set

$$\{E_{ij} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$$

spans $M_{m \times n}(\mathbb{R})$.

If a linear combination of the E_{ij} equals the zero matrix, then all coefficients must be zero; hence the set is linearly independent.

Since there are mn such matrices,

$$\boxed{\text{Basis} = \{E_{ij}\}}, \quad \boxed{\dim_{\mathbb{R}}(M_{m \times n}(\mathbb{R})) = mn}.$$

2. Find a basis and the dimension of the following subspaces W of the corresponding vector spaces.

(a) $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ (b) $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0, 2x + y + z = 0\}$

(c) $W = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : b = c = 0 \right\}$ (d) $W = \{A \in M_{2 \times 2}(\mathbb{R}) : A = A^T\}$

(e) $W = \{A \in M_{2 \times 2}(\mathbb{R}) : A = -A^T\}$ (f) $W = \{A \in M_{n \times n}(\mathbb{R}) : \text{trace}(A) = 0\}$

(g) Fix $a \in \mathbb{R}$, $W = \{f(x) \in P_n(\mathbb{R}) : f(a) = 0\}$ (h) $W = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : a + b = 0 \right\}$.

(i) $W = \{A = (a_{ij}) \in M_n(\mathbb{R}) \mid a_{ij} = 0 \text{ if } i \text{ is even}\}$ (j) $W = \{A \in M_{2 \times 2}(\mathbb{C}) : A = \bar{A}^T\}$ over $\mathbb{R}$.

(a) $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$.

From $x + y + z = 0$ we obtain $x = -y - z$. Thus

$$(x, y, z) = (-y - z, y, z) = y(-1, 1, 0) + z(-1, 0, 1).$$

Hence

$$W = \text{span}\{(-1, 1, 0), (-1, 0, 1)\}.$$

These vectors are linearly independent, so

$$\boxed{\text{Basis} = \{(-1, 1, 0), (-1, 0, 1)\}}, \quad \boxed{\dim W = 2}.$$

(b) $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0, 2x + y + z = 0\}$.

Subtracting the first equation from the second gives $x = 0$. Substituting into $x + y + z = 0$ yields $y = -z$. Thus

$$(x, y, z) = (0, -z, z) = z(0, -1, 1).$$

Therefore

$$\boxed{\text{Basis} = \{(0, -1, 1)\}}, \quad \boxed{\dim W = 1}.$$

(c) $W = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}(\mathbb{R}) : b = c = 0 \right\}$.

Such matrices have the form

$$\begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Hence a basis is

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}, \quad \boxed{\dim W = 2}.$$

- (d) $W = \{A \in M_{2 \times 2}(\mathbb{R}) : A = A^T\}$ (symmetric matrices).

Such matrices have the form

$$\begin{bmatrix} a & b \\ b & d \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Thus

$$\text{Basis} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}, \quad \boxed{\dim W = 3}.$$

- (e) $W = \{A \in M_{2 \times 2}(\mathbb{R}) : A = -A^T\}$ (skew-symmetric matrices).

Such matrices have the form

$$\begin{bmatrix} 0 & a \\ -a & 0 \end{bmatrix} = a \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Hence

$$\text{Basis} = \left\{ \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\}, \quad \boxed{\dim W = 1}.$$

- (f) $W = \{A \in M_{n \times n}(\mathbb{R}) : \text{trace}(A) = 0\}$.

Since $\dim M_{n \times n}(\mathbb{R}) = n^2$ and the trace condition imposes one linear constraint,

$$\boxed{\dim W = n^2 - 1}.$$

A basis consists of all E_{ij} ($i \neq j$) together with $E_{11} - E_{nn}, E_{22} - E_{nn}, \dots, E_{n-1, n-1} - E_{nn}$.

- (g) Fix $a \in \mathbb{R}$, $W = \{f(x) \in P_n(\mathbb{R}) : f(a) = 0\}$.

If $f(a) = 0$, then $(x - a)$ divides $f(x)$. Thus

$$W = \{(x - a)g(x) : g(x) \in P_{n-1}(\mathbb{R})\}.$$

Hence a basis is

$$\{(x - a), (x - a)x, \dots, (x - a)x^{n-1}\}, \quad \boxed{\dim W = n}.$$

- (h) $W = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a + b = 0 \right\}$.

Since $b = -a$, matrices have the form

$$\begin{bmatrix} a & -a \\ c & d \end{bmatrix} = a \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Thus

$$\text{Basis} = \left\{ \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}, \quad \boxed{\dim W = 3}.$$

- (i) $W = \{A = (a_{ij}) \in M_n(\mathbb{R}) : a_{ij} = 0 \text{ if } i \text{ is even}\}$.

Only odd-numbered rows may be nonzero. If $n = 2k$ is even, there are k odd rows; if $n = 2k + 1$, there are $k + 1$ odd rows.

Hence

$$\boxed{\dim W = (\text{number of odd rows}) \cdot n = \left\lceil \frac{n}{2} \right\rceil n}.$$

A basis consists of E_{ij} with i odd.

- (j) $W = \{A \in M_{2 \times 2}(\mathbb{C}) : A = \bar{A}^T\}$ over $\mathbb{R}$.

Such matrices are Hermitian:

$$A = \begin{bmatrix} a & z \\ \bar{z} & d \end{bmatrix},$$

where $a, d \in \mathbb{R}$ and $z \in \mathbb{C}$.

Write $z = x + iy$ with $x, y \in \mathbb{R}$. Then

$$A = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + x \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + y \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}.$$

Thus over $\mathbb{R}$,

$$\boxed{\text{Basis} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} \right\}}, \quad \boxed{\dim W = 4}.$$

3. Which of the following is a basis of $\mathbb{R}^3$ over $\mathbb{R}$?

- (a) $\{(1, 1, 0), (5, 5, 1), (8, 8, 4)\}$ (b) $\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$ (c) $\{(1, 1, 0), (0, 1, 1), (1, 0, 1), (0, 0, 1)\}$
 (d) $\{(1, 1, 0), (0, 0, 3)\}$.

4. Which of the following is a basis of $\mathbb{R}^3$ over $\mathbb{R}$?

- (a) $\{(1, 1, 0), (5, 5, 1), (8, 8, 4)\}$

Let

$$a(1, 1, 0) + b(5, 5, 1) + c(8, 8, 4) = (0, 0, 0).$$

This gives the system

$$a + 5b + 8c = 0, \quad a + 5b + 8c = 0, \quad b + 4c = 0.$$

The coefficient matrix is

$$A = \begin{pmatrix} 1 & 5 & 8 \\ 1 & 5 & 8 \\ 0 & 1 & 4 \end{pmatrix}.$$

Row reduce:

$$R_2 \rightarrow R_2 - R_1$$

$$\sim \begin{pmatrix} 1 & 5 & 8 \\ 0 & 0 & 0 \\ 0 & 1 & 4 \end{pmatrix}.$$

Interchange R_2 and R_3 :

$$\sim \begin{pmatrix} 1 & 5 & 8 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{pmatrix}.$$

The system becomes

$$a + 5b + 8c, \quad b + 4c = 0.$$

Since c is free, there are nontrivial solutions. Hence the vectors are linearly dependent.

Therefore the set is **not** a basis of $\mathbb{R}^3$.

- (b) $\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$

Let

$$a(1, 1, 0) + b(0, 1, 1) + c(1, 0, 1) = (0, 0, 0).$$

This gives

$$(a + c, a + b, b + c) = (0, 0, 0),$$

so the system is

$$a + c = 0, \quad a + b = 0, \quad b + c = 0.$$

The coefficient matrix is

$$B = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Row reduce:

$$R_2 \rightarrow R_2 - R_1$$

$$\sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix}.$$

$$R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Thus the only solution is

$$a = b = c = 0.$$

Hence the vectors are linearly independent. Since there are three independent vectors in $\mathbb{R}^3$ and $\dim(\mathbb{R}^3) = 3$, they form a basis.

- (c) $\{(1, 1, 0), (0, 1, 1), (1, 0, 1), (0, 0, 1)\}$

This set contains four vectors in $\mathbb{R}^3$. Since $\dim(\mathbb{R}^3) = 3$, any set of four vectors is linearly dependent. Hence it cannot be a basis.

- (d) $\{(1, 1, 0), (0, 0, 3)\}$

This set contains only two vectors. Since $\dim(\mathbb{R}^3) = 3$, at least three vectors are required to form a basis. Hence this set is not a basis.

5. For what real values of k , does the the set $\{(k, 0, 1), (1, k + 1, 1), (1, 1, 1)\}$ forms a basis of $\mathbb{R}^3$ over $\mathbb{R}$?
[Ans. $k \neq 0, 1$]

A set of three vectors in $\mathbb{R}^3$ forms a basis if and only if it is LI. From HW3, $k \neq 0, 1$.

6. Give three different basis for $M_{2 \times 2}(\mathbb{R})$ and $\mathbb{R}^2$ over $\mathbb{R}$.
7. Give three different bases for $M_{2 \times 2}(\mathbb{R})$ and $\mathbb{R}^2$ over $\mathbb{R}$.

(1) Bases for $M_{2 \times 2}(\mathbb{R})$.

Since $\dim M_{2 \times 2}(\mathbb{R}) = 4$, any set of four linearly independent matrices forms a basis.

Basis 1 (Standard basis).

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

Basis 2 (Symmetric and skew-symmetric decomposition).

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}.$$

Basis 3 (Another independent set).

$$\left\{ \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

Each of the above sets contains four linearly independent matrices, hence each forms a basis.

(2) Bases for $\mathbb{R}^2$.

Since $\dim(\mathbb{R}^2) = 2$, any two linearly independent vectors form a basis.

Basis 1 (Standard basis).

$$\{(1, 0), (0, 1)\}.$$

Basis 2.

$$\{(1, 1), (1, -1)\}.$$

Basis 3.

$$\{(2, 1), (1, 3)\}.$$

Each pair consists of two linearly independent vectors, hence each is a basis of $\mathbb{R}^2$.

8. Find all subspaces of $\mathbb{R}^2$ and $\mathbb{R}^3$.

9. Find all subspaces of $\mathbb{R}^2$ and $\mathbb{R}^3$.

(1) Subspaces of $\mathbb{R}^2$.

Let W be a subspace of $\mathbb{R}^2$. Since $\dim(\mathbb{R}^2) = 2$, the dimension of W can be 0, 1, or 2.

(a) $\dim W = 0$: The trivial subspace

$$W = \{(0, 0)\}.$$

(b) $\dim W = 1$: Any one-dimensional subspace is a line through the origin. That is,

$$W = \{tv : t \in \mathbb{R}\}, \quad v \in \mathbb{R}^2, v \neq 0.$$

(c) $\dim W = 2$: The whole space

$$W = \mathbb{R}^2.$$

(2) Subspaces of $\mathbb{R}^3$.

Let W be a subspace of $\mathbb{R}^3$. Since $\dim(\mathbb{R}^3) = 3$, the dimension of W can be 0, 1, 2, or 3.

(a) $\dim W = 0$: The trivial subspace

$$W = \{(0, 0, 0)\}.$$

(b) $\dim W = 1$: Any one-dimensional subspace is a line through the origin.

$$W = \{tv : t \in \mathbb{R}\}, \quad v \in \mathbb{R}^3, v \neq 0.$$

(c) $\dim W = 2$: Any two-dimensional subspace is a plane through the origin.

$$W = \{av_1 + bv_2 : a, b \in \mathbb{R}\}, \quad v_1, v_2 \text{ linearly independent.}$$

(d) $\dim W = 3$: The whole space

$$W = \mathbb{R}^3.$$

10. Using the row echelon form of a matrix, find the row rank, the column rank, the rank and the nullity of the following matrices. Also, find bases for the row space, the column space, and the null space of the matrices.

(a) $A = \begin{bmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 6 & 9 \\ 1 & 1 & 2 & 6 \end{bmatrix}$ (b) $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 1 & 0 \end{bmatrix}$ (c) $A = \begin{bmatrix} -2 & 0 & 0 & 3 \\ 1 & 5 & 3 & 0 \\ 3 & 2 & 1 & 6 \\ 3 & 5 & 3 & -3 \end{bmatrix}$ Answers: (a) 2 (b) 2 (c) 3.

(a) We have

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 6 & 9 \\ 1 & 1 & 2 & 6 \end{pmatrix}.$$

Step 1: Row reduce A to row echelon form.

Eliminate entries below first pivot (2 in row 1, column 1):

$$R_2 \rightarrow R_2 - \frac{3}{2}R_1, \quad R_3 \rightarrow R_3 - \frac{1}{2}R_1$$

$$\begin{pmatrix} 2 & 1 & 4 & 3 \\ 0 & \frac{1}{2} & 0 & \frac{9}{2} \\ 0 & \frac{1}{2} & 0 & \frac{9}{2} \end{pmatrix}$$

Eliminate duplicate row:

$$R_3 \rightarrow R_3 - R_2 \Rightarrow \begin{pmatrix} 2 & 1 & 4 & 3 \\ 0 & \frac{1}{2} & 0 & \frac{9}{2} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

This is row echelon form (REF).

Step 2: Row rank, column rank, rank, nullity

- Row rank = number of nonzero rows = 2,
- Column rank = rank = 2,
- Rank = 2,
- Nullity = $n - \text{rank} = 4 - 2 = 2$.

Step 3: Bases

(i) Row space:

Take the nonzero rows of REF as a basis :

$$\text{Row space basis} = \left\{ (2, 1, 4, 3), \left(0, \frac{1}{2}, 0, \frac{9}{2} \right) \right\}$$

(If desired, multiply second row by 2 to remove fractions: $(0, 1, 0, 9)$.)

(ii) Column space:

Pivot columns in REF are columns 1 and 2 $\rightarrow$ take corresponding columns from original matrix:

$$\text{Column space basis} = \left\{ \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\}$$

(iii) Null space:

Let free variables be $x_3 = s, x_4 = t$. Solve using REF:

From row 2:

$$\frac{1}{2}x_2 + \frac{9}{2}x_4 = 0 \implies x_2 + 9x_4 = 0 \implies x_2 = -9x_4 = -9t.$$

From row 1:

$$2x_1 + x_2 + 4x_3 + 3x_4 = 0 \implies 2x_1 + (-9t) + 4s + 3t = 0 \implies x_1 = -2s + 3t$$

Hence general solution:

$$(x_1, x_2, x_3, x_4) = s(-2, 0, 1, 0) + t(3, -9, 0, 1).$$

$$\text{Null space basis} = \{(-2, 0, 1, 0), (3, -9, 0, 1)\}.$$

Follow the above for (b) and (c).

11. Which of the following sets spans $\mathbb{R}^3$ over $\mathbb{R}$?

(a) $\{(1, 1, 0), (1, -1, 0), (0, 1, 1)\}$ (b) $\{(2, 1, 1), (1, 1, 1), (0, 2, 1), (0, 0, 3)\}$

(c) $\{(1, 1, 0), (0, 0, 1)\}$.

A set of vectors spans $\mathbb{R}^3$ if every vector in $\mathbb{R}^3$ can be written as a linear combination of the set. Equivalently, the set spans $\mathbb{R}^3$ if the vectors are linearly independent and there are at least 3 vectors (or if the rank of the matrix formed by the vectors is 3).

(a) $\{(1, 1, 0), (1, -1, 0), (0, 1, 1)\}$

Form the matrix with these vectors as rows:

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Check linear independence using row echelon form:

$$R_2 \rightarrow R_2 - R_1 \Rightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

$$R_3 \rightarrow R_3 + \frac{1}{2}R_2 \Rightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

All three rows are nonzero, thus, rank = 3.

Thus the set is linearly independent and contains 3 vectors. Therefore it spans $\mathbb{R}^3$.

(b) $\{(2, 1, 1), (1, 1, 1), (0, 2, 1), (0, 0, 3)\}$

There are 4 vectors in $\mathbb{R}^3$. Since $\dim(\mathbb{R}^3) = 3$, any set with more than 3 vectors is automatically linearly dependent. We only need 3 linearly independent vectors to span.

Form a 4×3 matrix with the vectors as rows:

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix}.$$

Perform row reduction to find the rank:

$$R_2 \rightarrow R_2 - \frac{1}{2}R_1,$$

$$\sim \begin{pmatrix} 2 & 1 & 1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix}.$$

$$R_3 \rightarrow R_3 - 4R_2 \Rightarrow \begin{pmatrix} 2 & 1 & 1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \\ 0 & 0 & 3 \end{pmatrix}.$$

$$R_4 \rightarrow R_4 + 3R_3 \Rightarrow \begin{pmatrix} 2 & 1 & 1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

The row echelon form has 3 nonzero rows. Hence the rank of the matrix is 3.

Since $\dim(\mathbb{R}^3) = 3$, these four vectors span $\mathbb{R}^3$ (even though they are linearly dependent).

(c) $\{(1, 1, 0), (0, 0, 1)\}$

There are only 2 vectors in $\mathbb{R}^3$. Since $\dim(\mathbb{R}^3) = 3$, 2 vectors cannot span the space.

Hence this set does **not** span $\mathbb{R}^3$.

12. Extend the set $\{(1, 2, 1, 2), (2, 1, 2, 1)\}$ to a basis of $\mathbb{R}^4$ over $\mathbb{R}$.

13. Extend the set

$$S = \{(1, 2, 1, 2), (2, 1, 2, 1)\}$$

to a basis of $\mathbb{R}^4$ over $\mathbb{R}$ using row echelon form.

Step 1: Form a matrix with the given vectors as rows.

$$A = \begin{pmatrix} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \end{pmatrix}.$$

Step 2: Row reduce to echelon form.

$$R_2 \rightarrow R_2 - 2R_1$$

$$\sim \begin{pmatrix} 1 & 2 & 1 & 2 \\ 0 & -3 & 0 & -3 \end{pmatrix}.$$

Now the matrix is in row echelon form.

Step 3: Identify free columns.

The pivot columns are columns 1 and 2. Columns 3 and 4 are free, we can use them to form new independent vectors.

$$v_3 = (0, 0, 1, 0), \quad v_4 = (0, 0, 0, 1)$$

These vectors correspond to the free variables and are linearly independent of the original two rows.

Step 4: Extend the set to a basis.

$$\boxed{\{(1, 2, 1, 2), (2, 1, 2, 1), (0, 0, 1, 0), (0, 0, 0, 1)\}}$$

This set has four vectors in $\mathbb{R}^4$ and is linearly independent, hence forms a basis.

14. Find the dimension of the solution space of the following set of homogeneous system of linear equations:
 $3x + 4y - z - 6w = 0, 2x + 3y + 2z - 3w = 0, 2x + y - 14z - 9w = 0, x + 3y + 13z + 3w = 0.$

Step 1: Form the augmented matrix.

$$A = \begin{pmatrix} 3 & 4 & -1 & -6 \\ 2 & 3 & 2 & -3 \\ 2 & 1 & -14 & -9 \\ 1 & 3 & 13 & 3 \end{pmatrix}$$

Step 2: Row reduce to echelon form.

Swap $R_1 \leftrightarrow R_4$:

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & -3 \\ 2 & 1 & -14 & -9 \\ 3 & 4 & -1 & -6 \end{pmatrix}$$

Eliminate first column below pivot:

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 2R_1, \quad R_4 \rightarrow R_4 - 3R_1$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 0 & -3 & -24 & -9 \\ 0 & -5 & -40 & -15 \\ 0 & -5 & -40 & -15 \end{pmatrix}$$

Simplify R_2 :

$$R_2 \rightarrow -\frac{1}{3}R_2 \Rightarrow \begin{pmatrix} 1 & 3 & 13 & 3 \\ 0 & 1 & 8 & 3 \\ 0 & -5 & -40 & -15 \\ 0 & -5 & -40 & -15 \end{pmatrix}$$

Eliminate column 2 below pivot:

$$R_3 \rightarrow R_3 + 5R_2, \quad R_4 \rightarrow R_4 + 5R_2$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 0 & 1 & 8 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Eliminate above pivot in column 2:

$$R_1 \rightarrow R_1 - 3R_2 \Rightarrow \begin{pmatrix} 1 & 0 & -11 & -6 \\ 0 & 1 & 8 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Step 3: Identify pivot and free columns.

- Pivot columns: 1 and 2 - Free columns: 3 and 4

Step 4: Dimension of solution space.

$$\dim(\text{solution space}) = \text{number of free variables} = 4 - \text{Rank} = 2.$$

Step 5: General solution

Let $z = s, w = t$. Then

$$y = -8s - 3t, \quad x = 11s + 6t, \quad z = s, \quad w = t$$

$$(x, y, z, w) = s(11, -8, 1, 0) + t(6, -3, 0, 1)$$

$$\boxed{\text{Basis} = \{(11, -8, 1, 0), (6, -3, 0, 1)\}}, \quad \boxed{\dim = 2}.$$